

FOOD DELIVERY ANALYTICS

Delivery Time Prediction & Operational Insights

An Interactive Food Delivery Analytics & Prediction Platform

45,584 Delivery Records	26.3 min Average Delivery	0.8059 R ² Score
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IBM SkillsBuild — Data Analytics with AI

Project Report • 2026

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Computer Science & Engineering

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FOOD DELIVERY ANALYTICS AND DELIVERY TIME PREDICTION

Project Report

Submitted as part of	IBM SkillsBuild Data Analytics with AI Internship
Student Name	Lahari Koram
Department	Computer Science and Engineering
Institution	University College Of Engineering and Technology for Women
Academic Year	2026
Project Area	Data Analytics • Machine Learning • AI

Project focus

A compact end-to-end application that combines historical food-delivery analysis, delivery-time prediction and AI-assisted explanation in an interactive Streamlit dashboard.

CERTIFICATE

This is to certify that **Lahari Koram**, a student of the Department of Computer Science and Engineering, has completed the project titled “**Food Delivery Analytics and Delivery Time Prediction**” as part of the IBM SkillsBuild Data Analytics with AI project/internship work.

The project demonstrates the practical application of data preparation, exploratory data analysis, visualization, machine learning, interactive dashboard development and AI-assisted interpretation.

DECLARATION

I hereby declare that the project titled “**Food Delivery Analytics and Delivery Time Prediction**” is my work completed as part of the IBM SkillsBuild Data Analytics with AI Internship.

The project was developed for educational and analytical purposes using a food delivery dataset. The analysis, visualizations, machine learning model and interactive dashboard were developed as part of this project.

I confirm that the information presented in this report is accurate to the best of my knowledge.

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Date: 24-09-2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me in completing this project.

I am thankful to the **IBM SkillsBuild Data Analytics with AI Internship** for providing an opportunity to apply data analytics, machine learning and AI concepts to a practical project.

I also thank my mentors, friends and colleagues for their guidance, feedback and support throughout the development of this work.

This project provided an opportunity to connect classroom concepts with a practical data-science workflow involving data preparation, visualization, predictive modelling and interactive application development.

Thank You

ABSTRACT

Food delivery platforms generate large amounts of data related to orders, delivery personnel, restaurants, locations, traffic, weather, vehicles and delivery times. Analysing this information can help identify delivery patterns and support delivery-time estimation.

This project analyses **45,584 food delivery records from India covering February to April 2022**. The workflow combines data preparation, exploratory data analysis, feature engineering, machine learning, visualization and interactive dashboard development.

A **Random Forest regression model** was developed to estimate delivery time. The displayed evaluation reports a **Mean Absolute Error of 3.265 minutes** and an **R² score of 0.8059** on the test data.

The final Streamlit application provides an overview, detailed delivery analytics, prediction functionality and methodology/insight views. An AI-assisted explanation layer is also included to make analytical results easier to understand.

Keywords: Data Analytics, Food Delivery, Machine Learning, Random Forest, Regression, Streamlit, Artificial Intelligence

INTRODUCTION & OBJECTIVES

Food delivery time can vary because of distance, traffic, weather, vehicle condition, workload, order characteristics and other operational factors. This project studies these variables using historical delivery data and combines analytics with machine learning to estimate delivery duration.

Objectives

- Analyse historical food-delivery records.
- Identify patterns in delivery time and operational conditions.
- Compare delivery performance across traffic, weather, vehicle, order and city categories.
- Engineer useful features for predictive modelling.
- Develop a regression model for delivery-time estimation.
- Evaluate the model using standard regression metrics.
- Build an interactive Streamlit dashboard.
- Use AI-assisted explanations to simplify analytical results.

Project Workflow

DATA → CLEANING → EDA → FEATURE ENGINEERING → MACHINE LEARNING
→ EVALUATION → DASHBOARD → AI EXPLANATION

DATASET & DATA ANALYSIS

Dataset Item	Details
Records	45,584
Location	India
Period	February – April 2022
Target	Delivery Time
Format	Structured / Tabular

Important Variables

Delivery Person Age • Delivery Person Rating • Restaurant Location • Delivery Location • Order Date • Weather • Traffic • Vehicle Condition • Type of Order • Type of Vehicle • Festival • City Type • Multiple Deliveries • Delivery Time

Analysis Areas

Delivery-time distribution • Traffic analysis • Weather analysis • Vehicle analysis • Order-type analysis • City-type analysis • Simultaneous-delivery analysis • Feature importance

Suggested visuals for the final version

Insert 2–3 actual screenshots/charts from your Streamlit dashboard here, with short captions such as “Delivery Time Distribution” and “Delivery Analytics by Category.”

MACHINE LEARNING METHODOLOGY

The prediction task is formulated as a supervised regression problem. The target variable is delivery time in minutes.

Model: Random Forest Regression

Random Forest combines multiple decision trees to produce a prediction. It is suitable for this project because it can model nonlinear relationships between different delivery and operational features.

Model Workflow

Clean Data → Select Features → Encode Categories → Train/Test Split → Train Random Forest → Predict → Evaluate

Model Results

Metric	Result	Interpretation
MAE	3.265 minutes	Average absolute prediction error
R ²	0.8059	Variation represented by the model on the test data

Important features shown by the model include delivery distance, rider age, vehicle condition, festival period, city type and order hour. Feature importance describes model behaviour and should not be interpreted as proof of causation.

DASHBOARD & AI INTEGRATION

Interactive Streamlit Dashboard

Section	Purpose
Overview	High-level delivery statistics and distributions
Delivery Analytics	Detailed comparisons across operational categories
Prediction	Estimated delivery time using the trained model
Insights & Methodology	Model performance, feature importance and explanation

AI Integration

AI is used as an explanation layer. The Random Forest model produces the numerical prediction, while the AI component can help convert model-related information and analytical results into natural-language explanations.

The AI layer should not be treated as the source of the numerical prediction, and API credentials should never be included in the report.

Add 1–2 actual screenshots of your Streamlit dashboard here.

RESULTS & KEY FINDINGS

Result	Value
Dataset size	45,584 records
Average delivery time	≈ 26.3 minutes
Random Forest MAE	3.265 minutes
Random Forest R ²	0.8059

Key Findings

- Delivery distance is one of the relatively important features used by the model.
- Rider age and vehicle condition also contribute to model predictions.
- Traffic and weather provide useful categories for analysing delivery performance.
- Simultaneous deliveries can be examined as a workload-related factor.
- Delivery patterns can be compared across vehicle, order and city categories.

Note: Model metrics describe performance on the available test data and do not guarantee the same accuracy on future real-world data.

CONCLUSION & FUTURE SCOPE

Conclusion

The Food Delivery Analytics and Delivery Time Prediction project demonstrates an end-to-end application of data analytics, machine learning, visualization and AI. Historical delivery data was transformed into an interactive application capable of exploring delivery patterns and estimating delivery duration.

The Random Forest model achieved an MAE of 3.265 minutes and an R^2 score of 0.8059 on the test data. The Streamlit dashboard provides a practical interface for viewing analytics, predictions and methodology.

Future Scope

- Integrate real-time traffic and weather information.
- Add geospatial maps and route-level analysis.
- Compare additional machine-learning algorithms.
- Support real-time prediction using live order information.
- Expand AI-generated analytical summaries and recommendations.
- Retrain and monitor the model as new data becomes available.

Technologies

Python • Pandas • NumPy • Scikit-learn • Streamlit • Plotly • Random Forest • AI API

Thank you

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